

1. 🏢 Interactive First-Person Digital Twin

This is your **wow factor**.

Judge selects a building → enters it → moves through floors.



Click any apartment/space → show its **3D ULPIN, area, floor, ownership/property attributes, etc.**

2. 🧠 AI Automatically Understands the Building

Don't just reconstruct geometry.

Understand it.

From the point cloud / mesh:

Raw 3D Scan



AI





AI detects:

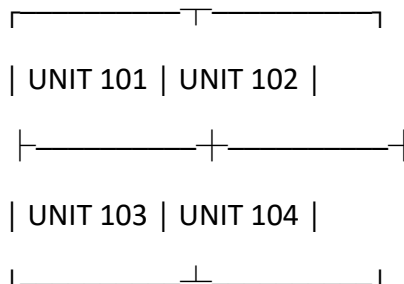
- Buildings
- Floors
- Rooms
- Walls
- Doors
- Windows
- Staircases
- Parking
- Common areas
- Vertical shafts

This directly strengthens the **automated building extraction + floor segmentation** part of the PS.

3. 🔥 Automatic 3D Property Parcel Creation

This is where your project becomes **much more than a 3D viewer**.

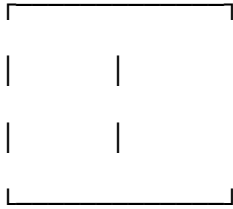
Suppose a floor contains:



AI identifies the boundaries.

Then creates actual **3D volumes**:

UNIT 102



ULPIN

Now you're implementing **vertical parcel delineation**, one of the core requirements.

4. 3D ULPIN That Actually Represents a Volume

This is potentially your strongest conceptual feature.

Traditional:

ULPIN



2D parcel

Your system:

3D ULPIN



(X,Y,Z)

+

Floor

+

Vertical extent

+

Property volume

For example:

ULPIN: IN-TN-XXXX-XXXX

Geometry:

X: ...

Y: ...

Z: Floor 3 → Floor 4

Volume:

Apartment 302

So two properties can occupy the **same X-Y footprint but different Z ranges**.

That's exactly the type of problem the PS is targeting.

5. 📡 Change Detection

This could be **VERY powerful for government**.

Scan the same building again after 6 months.

SCAN 2026

+

SCAN 2027

↓

AI CHANGE DETECTION

↓

┌──────────────────┐

| ⚠️ New construction |

| ⚠️ Modified room |

| ⚠️ Added floor |

| ⚠ Boundary change |

└──────────────────┘

You could visualize:

Before vs After

This turns your platform from a **one-time mapping system** into an **ongoing property monitoring system**.

6. ⚠ AI Topology & Cadastral Validation

The PS explicitly mentions **intelligent topology validation**.

So build an AI validation engine.

For example:

Property A

|

└─ overlaps Property B ❌

|

└─ outside parent parcel ❌

|

└─ missing floor ❌

|

└─ valid geometry ✅

The system could automatically flag:

- Overlapping properties
- Gaps
- Invalid boundaries
- Floor inconsistencies
- Properties outside parent parcels

- Duplicate spatial identities
- Impossible vertical relationships

This is **far more relevant to the PS than simply making a beautiful 3D model.**

7. 🌐 2D + 3D + Data in ONE Interface

Make three modes:

🗺️ 2D Cadastral

Traditional parcel map.

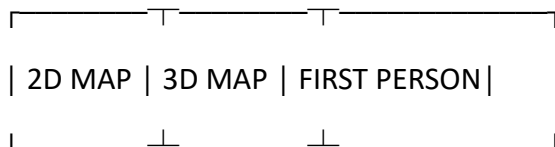
🏠 3D Cadastral

Building + floors + property volumes.

🎮 First-person

Actually walk inside the building.

GOVERNMENT 3D CADASTRE



Click a parcel in **2D** → corresponding 3D building highlights.

Click an apartment in **3D** → corresponding cadastral parcel highlights.

Walk inside → click a property → its ULPIN appears.

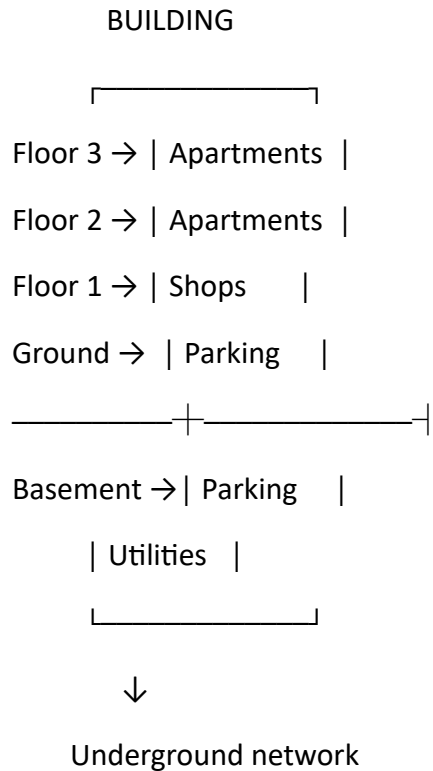
That integration is the killer feature.

🤖 And then add the underground layer

The PS specifically mentions **underground infrastructure.**

Don't ignore it.

Your 3D environment could have layers:



Toggle:



Properties



Underground utilities



Infrastructure



Terrain



Cadastral parcels

Now you've got an actual **volumetric cadastral viewer**.

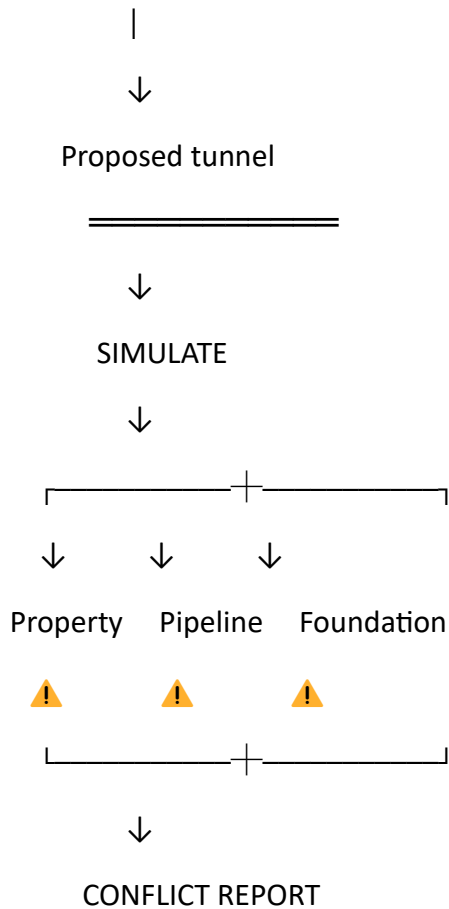
1. Underground Infrastructure Simulation

This is a very good addition.

Suppose a new underground tunnel is proposed.

Instead of digging first and discovering conflicts later:

EXISTING 3D CITY



The system can show:

Proposed Tunnel T-01 intersects underground utility U-17 at 3.2 m depth.

or:

Proposed infrastructure passes below Property X.

This is directly relevant to the PS because it explicitly includes **underground infrastructure** and infrastructure planning.

Even better:

Allow an authorized user to **draw/propose a future tunnel** inside the 3D environment.

Then your system performs:

3D collision detection → affected properties → affected utilities → clearance analysis → impact report.

That's a legitimate engineering feature, not just visualization.

2. Public vs Government Access

This is also excellent, but I'd frame it as **role-based spatial access control**.

Public

They can see:

 Roads

 Public spaces

 Building exterior

 Public facilities

But private property interiors are hidden/restricted.

Government-authorized user

Can access:

3D property boundaries

Floor plans

Property identifiers

Infrastructure

Underground layers

Inspection information

Property owner

Can access:

Their property

Their 3D unit

Property documents

Property boundaries

Inspector

Can access:

Assigned properties

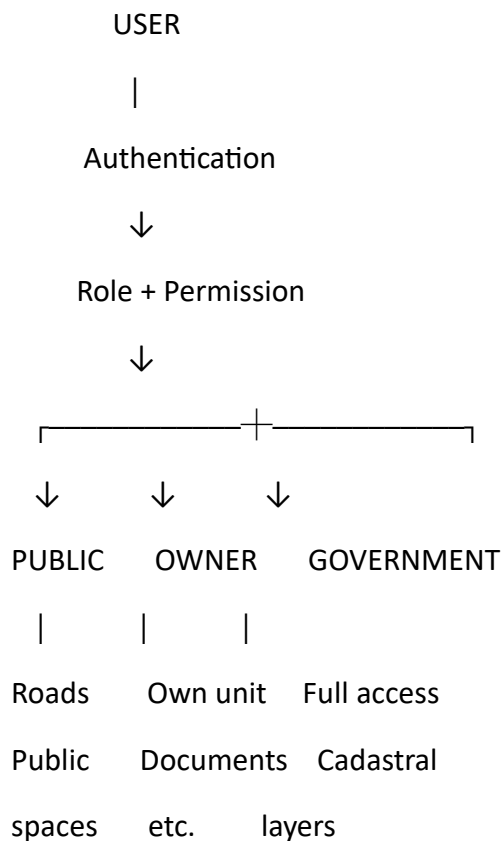
Inspection mode

3D measurement

Changes

Validation results

So:

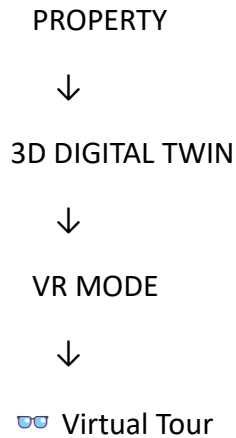


Important: don't claim "government permission" as a real legal authorization mechanism unless your prototype is actually integrated with an authority. Call it a **prototype authorization workflow**.

3. 🧐 VR Property Exploration

This is your **wow factor**, especially for property visualization.

Imagine someone wants to explore an apartment before visiting:



They can:

- Walk through rooms
- Look around
- See dimensions
- Explore floor layouts
- View amenities
- Compare units

But keep this separate from cadastral authority.

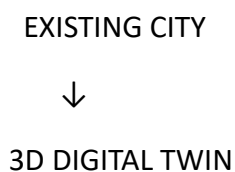
The VR experience is primarily **visualization**, while the cadastral system remains the authoritative spatial/data layer.

4. 🔥 Add a "Future Development Simulator"

This could be one of your strongest features.

Imagine:

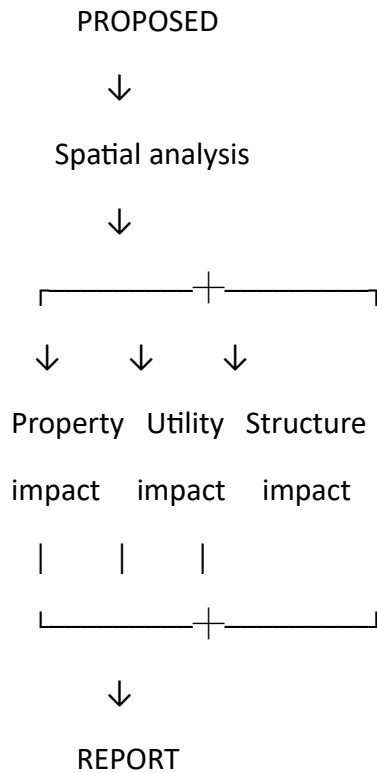
Current state



Proposed project

- + NEW TUNNEL
- + NEW ROAD
- + NEW BUILDING
- + NEW UTILITY

Simulate



The government can ask:

"If we construct this tunnel here, what existing assets will be affected?"

That's a much more powerful use of the 3D cadastral database.